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# Lecture 10

## ECE 4107 (Section A )

### Telecommunication Engineering



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## DISTRIBUTED SPC

- Control functions shared between many processors within the exchange.
- Low cost microprocessors offer better availability and reliability than centralised SPC.
- Exchange control functions decomposed horizontally or vertically.



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### DISTRIBUTED SPC - Vertical decomposition

- Exchange divided into blocks.
- Each block assigned to a processor.
- Performs all control functions related to that block of equipments.
- Total control system consists of several control units coupled together.
- Processor in each block may be duplicated for redundancy.
- Operates in any of three dual processor modes as explained earlier.
- Modular so that more can be added when exchange is expanded.

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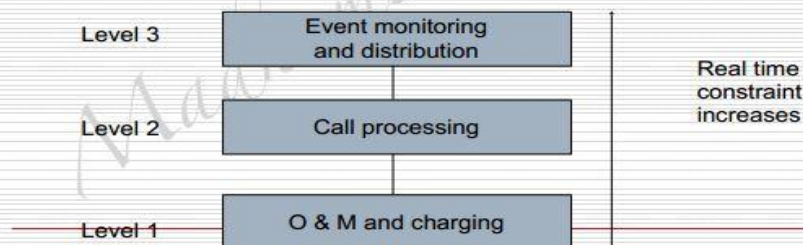


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### DISTRIBUTED SPC - Horizontal decomposition

- Each processor performs one or some of exchange control functions.
- Chain of processors for 3 functions.
- Entire chain may duplicate for redundancy.

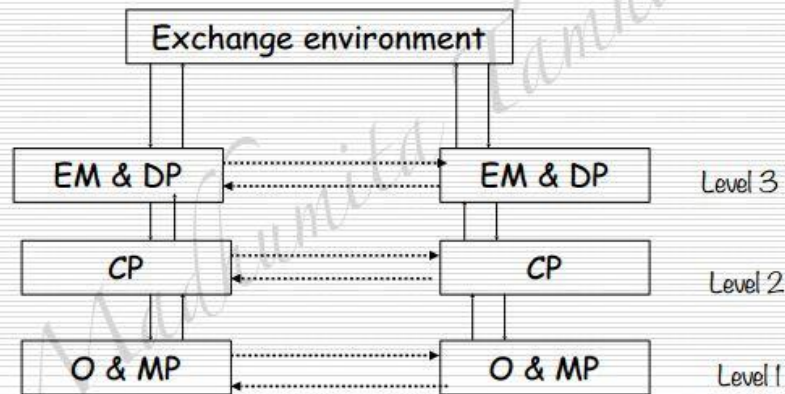


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### DISTRIBUTED SPC - Horizontal decomposition

- Entire chain may duplicate for redundancy.



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### DISTRIBUTED SPC - Level 3 processing

- Handles scanning, distribution and marking functions.
- Operations simple, specialised and well defined.
- Sets or senses binary conditions in F/F or registers.
- Achieves control by sensing or altering binary conditions using CONTROL WORD
- Hard wired or micro programmed device.
- Compare micro programmed control to Hard wired control.



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## Comparison of Micro Program and Hard-wired Control

| Micro program Control                          | Hard-wired Control                                  |
|------------------------------------------------|-----------------------------------------------------|
| Flexible                                       | Not Flexible                                        |
| Slow                                           | Fast                                                |
| More expensive for moderate processing         | Less expensive for moderate processing              |
| Easy to implement complex processing functions | Difficult to implement complex processing functions |
| Introducing new service is easy                | Not easily possible                                 |
| Easy to maintain                               | Difficult to maintain                               |



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### DISTRIBUTED SPC - Level 3 processing

- Set of control words stored in memory and read one by one.
- **Horizontal control** - One bit per every control signal.
- Flexible and fast.
- Expensive as large width - depends on number of signals.
- **Vertical control** - Each signals binary encoded as a word.
- Time too large as at a time only one signal.
- Mid approach chosen.
- Control word contains group of encoded words.

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## DISTRIBUTED SPC - Level 2 processing

- Processors for call processing.
- Called switching processors.
- Instructions designed to allow data to be packed more tightly in memory without increasing access time.
- Processor designed to ensure over 99.9% availability, fault tolerance and security of operation.
- I/O data transfer order of 100 kilobytes per s.
- I/O technique:
  - Program controlled data transfer .
  - Direct memory access.



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## DISTRIBUTED SPC - Level 2 processing

- Traffic handling capacity of control equipment limited by capacity of switching processor.
- Load on switching processor measured by occupancy  $t$ .
- Occupancy: Fraction of unit time for which processor is occupied.
- $t = a + bN$
- $a$  = fixed overhead depending on exchange capacity and configuration
- $b$  = average time to process one call.
- $N$  = number of calls per unit time.



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## DISTRIBUTED SPC - Level 2 processing

- ▣ a depends on scanning workload which depends on number of subscriber lines, trunks and service circuits in exchange.)
  - a estimated by knowing total lines, instructions required to scan one line and average execution time per instruction.
- ▣ b depends on type of call process.
  - Incoming call process time less than outgoing or transit calls etc..
  - Results of party busy or no answer etc also affect.
  - Type of subscribers (DTMF/rotary dial) also affect as grouping PBX lines change.



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## Nominal Load (number of call / unit time)

$$N_N = \frac{0.95 - a}{1.4 b}$$

- Usually the switch processor is designed to handle a traffic load which is **40%** higher than the nominal load
- At overload situation switch processors may be loaded 95% of its capacity



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### DISTRIBUTED SPC - Level 1 processing (O&M)

- Administer the exchange H/W and S/W.
- Add, modify and delete information in translation table.
- Change subscriber class of service.
- Put a new trunk or line into operation.
- Supervise operation of the exchange.
- Monitor traffic.
- Detect and locate fault and errors.
- Run diagnostic and test programs.
- Man-machine interaction.

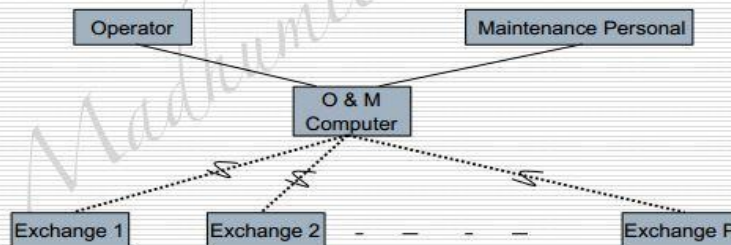


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### DISTRIBUTED SPC - Level 1 processing (O&M)

- Less subject to real time constraint.
- Less need for concurrent processing.
- Single O&M computer caters to many exchanges.
- Helps diagnosis of many from one location.

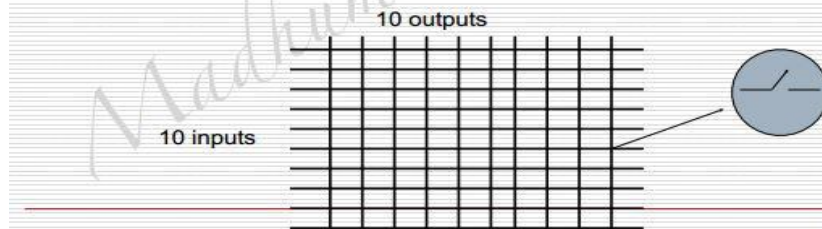


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### SINGLE STAGE NETWORKS

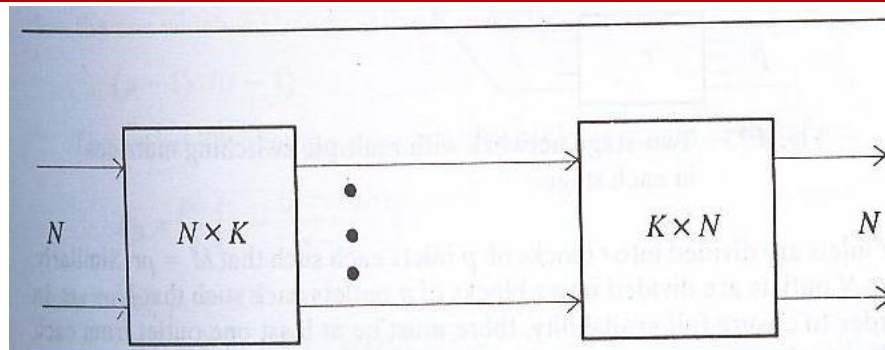
- No of cross points will be  $10 \times 10 = 100$ .
- Fully connected so no blocking.
- Used for 10-25% time on average.
- Remains idle. Waste of infrastructure.



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### Two stage representation of $N \times N$ network



- For any single stage network there exists an equivalent multistage network
- So,  $N \times K$  single stage network with capacity  $K$  can be realized by a two stage network of  $N \times K$  and  $K \times N$  stages



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## Connections

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- Any of the  $N$  inlets can be connected to any of the  $K$  outputs of 1<sup>st</sup> stage
- Similarly, Any of the  $K$  inputs can be connected to any of the  $N$  outputs of 2<sup>nd</sup> stage
- So, there are  $K$  alternative paths and  $2NK$  switching elements

Any of the  $N$  inlets can be connected to any of the  $N$  outlets



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- Each stage has  $NK$  switching elements
  - Assume only a fraction of the subscribers to be active on an average
  - $K$  can be equal to  $N/16$
  - So, no. of switching elements,  $S = 2NK = N^2/8$
  - Example:
  - $N = 1024, K = 64$
  - $S = 131,027$
  - So, for large  $N$ , the switching matrix  $N \times K$  may still be difficult to realize practically

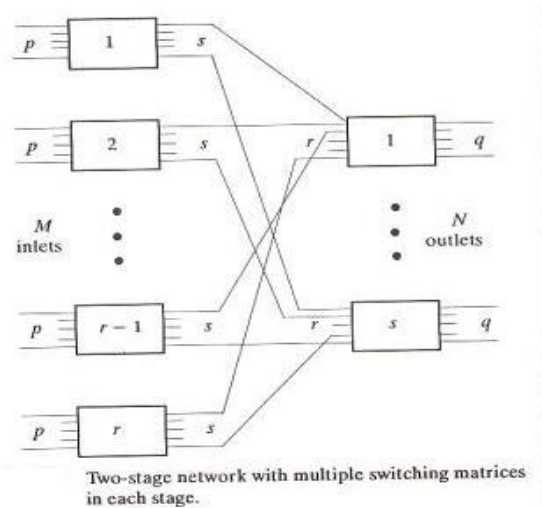


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## 2-stage Switching Network

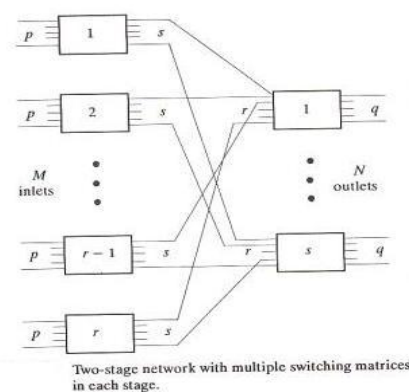
- $M$  inlets are divided into  $r$  blocks of  $p$  inlets.  $M = pr$
- $N$  outlets are divided into  $s$  blocks of  $q$  outlets.  $N = qs$



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- For full connectivity there must be at least one outlet from each block in the 1<sup>st</sup> stage terminating as inlet on every block of the 2<sup>nd</sup> stage
- So, **block sizes** are  $p \times s$  and  $r \times q$  respectively
- So,  $S = psr + qrs$
- Putting values for  $M, N$
- $S = Ms + Nr$        $M = pr, N = qs$
- $SC = rs$



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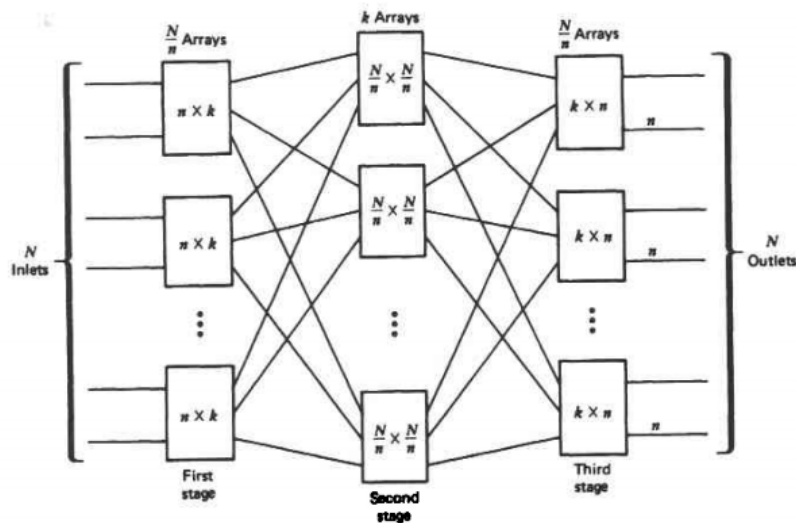
- The use of computers to control the switching functions of a central office led to the designation "electronic" switching
- Electronic switching system (ESS) or electronic automatic exchange (EAX)
- However, the switching matrices of these first-generation electronic switches are actually electromechanical in nature
- The first digital switches No. 4 ESS-digital toll switch



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### Three Stage Network



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## Number of Cross point of Three Stage Network

The total number of crosspoints  $N_X$  required by a three-stage switch, as shown in Figure 5.6, is

$$N_X = 2Nk + k \left( \frac{N}{n} \right)^2 \quad (5.1)$$

where  $N$  = number of inlets–outlets

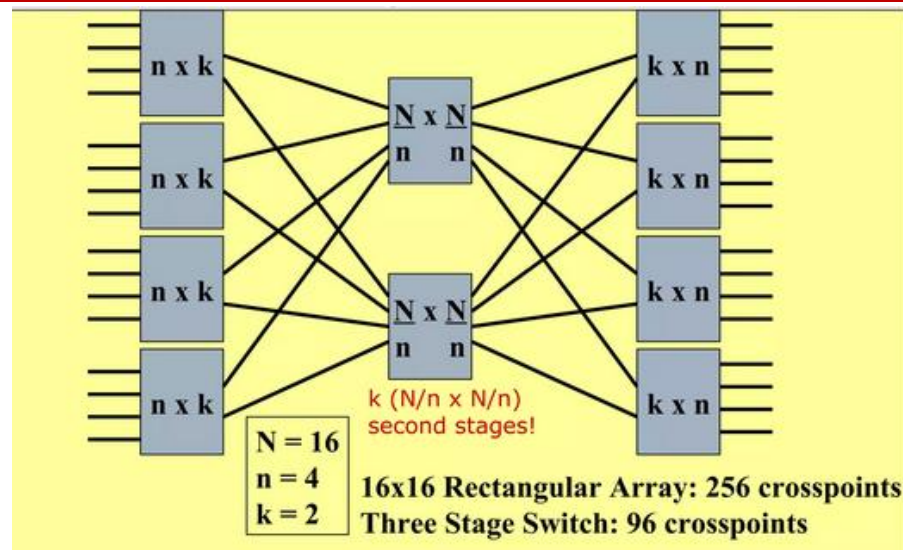
$n$  = size of each inlet–outlet group

$k$  = number of center-stage arrays



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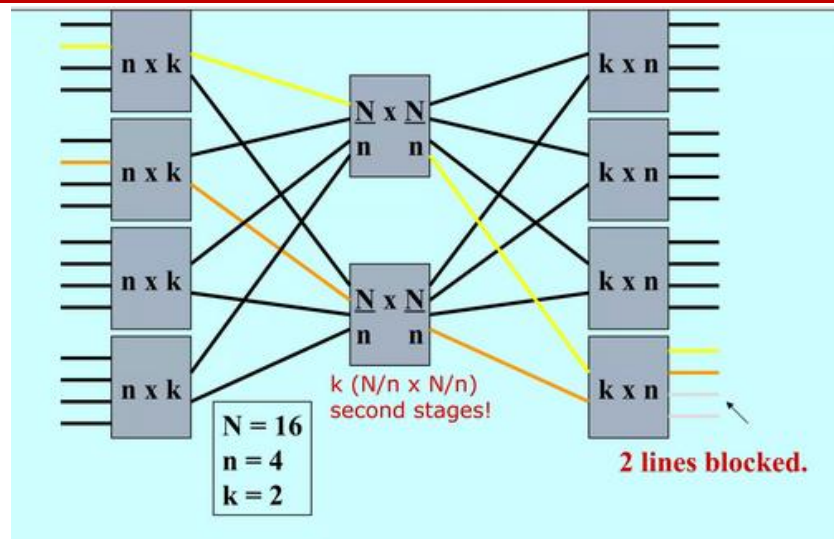
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### Nonblocking Switches

One attractive feature of a single-stage switch is that it is strictly nonblocking. If the called party is idle, the desired connection can always be established by selecting the particular crosspoint dedicated to the particular input-output pair. When crosspoints are shared, however, the possibility of blocking arises. In 1953 Charles Clos [2] of Bell Laboratories published an analysis of three-stage switching networks showing how many center-stage arrays are required to provide a strictly nonblocking operation. His result demonstrated that if each individual array is nonblocking, and if the number of center stages  $k$  is equal to  $2n - 1$ , the switch is strictly nonblocking.



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